# Using the MagicSchool AI Tutor

Algebra I · Module 2, Topic 2 · Lessons 1–3 · Use this to study for the Topic 2 Test

Name _____ Period _____ Date _____

**How to use it**

1. Open the MagicSchool link your teacher posted in Schoology.
2. Pick **one** question below. Type it in, or copy and paste it.
3. Read the answer. If it's confusing, use a **Helper phrase**.
4. Try a problem yourself, then ask the tutor to check your work.
5. Check the box ☐ next to each question you finish.

**Helper phrases** (type these any time)

- **"Explain it in shorter steps."**
- **"Use simpler words, please."**
- **"Show me an example first, then let me try one."**
- **"Give me a hint, not the answer."**
- **"Check my work: [type your steps]."**

## Lesson 1 · Transforming Linear Functions

☐ If f(x) = x and g(x) = 3·f(x), what happens to the graph? Show me with a table.

☐ What's the difference between g(x) = 2f(x) and g(x) = f(2x) when f(x) = x?

☐ How do I find the value of a in m(x) = a·f(x) from a graph?

☐ When does multiplying by a make a line steeper or flatter? What if a is negative?

☐ Let f(x) = x + 2. Are 3f(x) and f(3x) the same function? Go step by step.

## Lesson 2 · Vertical and Horizontal Transformations

☐ Why does f(x − 2) move the graph right and not left?

☐ What's the difference between f(x) + 4 and f(x + 4)?

☐ Here's a table for f(x). How do I shift it 2 units right and find the new y-intercept?

☐ Does a horizontal translation change the x-values or the y-values in a table?

☐ A food truck charges $50 plus $12 per hour, for at most 8 hours. How do I find the domain and range?

## Lesson 3 · Slopes of Perpendicular Lines

- ☐ How do I find the slope of a line perpendicular to $y = (3/4)x - 2$?
- ☐ What does "negative reciprocal" mean? Give me 3 examples to try.
- ☐ How do I write the equation of a line through (4, 0) that's parallel to $y = -1.5x + 2.4$?
- ☐ A line crosses the x-axis at 6. What point is that, and how do I use it in an equation?
- ☐ How do I write point-slope form from a graph?
- ☐ Why is the slope of $x = 5$ undefined, but the slope of $y = 5$ is zero?
- ☐ How can I tell if two lines are parallel, perpendicular, or neither?

## Review: all three lessons

- ☐ “Quiz me with 5 questions like the Topic 2 test. Give me one at a time and wait for my answer before you explain.”
- ☐ “I got this one wrong: [paste the problem]. Don't give me the answer. Give me a hint first.”

**Remember**

- The tutor helps you **think**. It is not for copying answers.
- One question at a time. Take a break when you need one.
- Still stuck? Write your question down and bring it to class.

**One thing I learned:**

**One question I still have:**